

Question	Marking details	Marks available					
		AO1	AO2	AO3	Total	Maths	Prac
7.	Indicative content Section A: Environmental impacts • (Excess) use of (inorganic) fertiliser • {Nitrates/ fertilizer} {washed/ leached/ run off} into {river/ water} • Causing algal blooms {along banks of Mississippi/ shown in image 7.1} • Blocking sunlight {causing plants to die/ means plants cannot photosynthesise} • Bacteria decompose plants using up oxygen • causing {lack of oxygen environment and {fish/ organisms} to die/ dead zones in {image 7.2/ named location} Section B: Root nodules • GM plants grew much better in low nitrate • Root nodules contain {N fixing bacteria/ rhizobium} • Correct reference to symbiosis/ mutualism • ({Rhizobium/ bacteria} in root nodules was) providing {nitrogen containing compounds/ nitrates/ ammonium} Section C: Lessening impact • Use of node forming plants increases {N content of soil/ soil fertility} • {Less/ no} (inorganic) fertiliser needed • Less {algal blooms/dead zones/ eutrophication} • Higher yield therefore less land needed for crops • Therefore more land available {for habitats/ increasing biodiversity}						

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				7-9 marks Indicative content of this level is. Detailed information from all three lists <i>The candidate constructs an articulate, integrated account, correctly linking relevant points, such as those in the indicative content, which shows sequential reasoning. The answer fully addresses the question with no irrelevant inclusions or significant omissions. The candidate uses scientific conventions and vocabulary appropriately and accurately.</i> 4-6 marks Indicative content of this level is detailed information from 2 of the lists or less detail from three areas. <i>The candidate constructs an account correctly linking some relevant points, such as those in the indicative content, showing some reasoning. The answer addresses the question with some omissions. The candidate usually uses scientific conventions and vocabulary appropriately and accurately.</i> 1-3 marks Indicative content of this level is some information from at least one of the lists to little information from more than one list. <i>The candidate makes some relevant points, such as those in the indicative content, showing limited reasoning. The answer addresses the question with significant omissions. The candidate has limited use of scientific conventions and vocabulary.</i> 0 marks <i>The candidate does not make any attempt or give a relevant answer worthy of credit.</i>						
				Question 7 total	0	5	4	9	0	0